Determinants of the Early-Exercise Premium for American Puts

What determines the premium of American puts over European puts in the SPDR S&P 500 ETF Options Chain.

Jacob Waldo, Siddharth Papani, Alex Xie

University of Southern California

2025

Contents

1 Introduction **2**

2 Theoretical Analysis **3**

3 Empirical Analysis and Results **4**

3.1 Construction of the Dependent Variable 4

3.1.1 Data Collection for Estimating $\boldsymbol{EEP_P}$ 4

3.1.2 Calculation of Present Value of Expected Dividends and Strike . . . 5

3.1.3 Estimating Theoretical EEP$_{\boldsymbol{P}}$ 5

3.1.4 Estimating Practical EEP$_{\boldsymbol{P}}$ 7

3.1.5 Automation of Bound Calculations 7

3.1.6 EEP$_{\boldsymbol{P}}$ Data Limitations 8

3.2 Monte Carlo Approach 9

3.2.1 Results 10

3.2.2 Interpretation of Coefficients 11

4 Conclusion **12**

1 Introduction

This paper examines the Early Exercise Premium (EEP) of American-style put options, with the central goal of addressing the following question: What factors determine the value of the early exercise premium in American puts, and how can this premium be reliably estimated under real market conditions when the premium itself is unobservable?

The early exercise premium represents the additional value an American put holds relative to its European counterpart. While American options allow exercise at any point prior to maturity, European options can only be exercised at expiration. This flexibility makes American options more valuable in theory, but the premium cannot be directly observed in derivative markets. Because no exact European counterparts exist on exchanges with the same underlying, strike, and maturity, the premium must instead be inferred indirectly. Developing reliable estimates is essential to evaluate how EEP responds to core option pricing variables and to assess whether market frictions prevent theoretical arbitrage conditions from holding in practice.

Our research uses the EEP as a lens into how markets operate beyond the idealized assumptions of standard models. Specifically, we investigate the role of time-to-maturity, implied volatility, and market frictions (including dividends, bid–ask spreads, and transaction costs) in shaping the premium. To ensure consistency with the underlying, implied volatility is measured using Bloomberg's 3-month at-the-money implied volatility index for SPY options, rather than broader benchmarks such as VIX. This choice ensures that the volatility measure is aligned with the same asset and contracts from which the EEP is derived, reducing mismatched inputs. At the same time, using an implied volatility at the index level rather than the implied volatilities of the specific options, we minimize the risk of introducing strong endogeneity into the regression.

A key challenge is that the EEP can only be bounded between theoretical upper and lower limits. To address this, we employ a Monte Carlo simulation framework, drawing repeatedly between bounds to generate latent realizations of ln(EEP). By doing so, we evaluate the robustness of regression estimates across thousands of simulated paths. The results confirm theory-consistent patterns: EEP decreases as time-to-maturity shortens and increases with implied volatility, even when measurement uncertainty is explicitly accounted for.

Using Bloomberg's end-of-day financial data, we offer a practical perspective on a concept that is central in theory but challenging to measure in practice. Our approach confirms option pricing theory in the presence of real market frictions and demonstrates how simulation can be utilized when key variables are only indirectly observable.

The sections outline the theoretical framework motivating our analysis, describe the data and Monte Carlo methodology used, and interpret the empirical results in the context of market behavior and theory.

2 Theoretical Analysis

To estimate the early exercise premium on American puts (EEP$_P$), we use the no-arbitrage condition embedded in put-call parity. For a dividend-paying stock, the European put-call parity relationship is defined as:

$$P_E(S_0', T, X) + S_0' = C_E(S_0', T, X) + X \cdot \exp(-r_e T)$$

This relationship enables us to calculate the theoretical value of a European put (P_E), which forms the standard against which we define the American EEP$_P$:

$$EEP_P = P_A - P_E$$

To estimate EEP$_P$ in practice, we use the prices of American puts and calls (P_A & C_A), discount the strike and expected dividends, and adjust for transaction costs arising from bid-ask spreads. Later in the paper, we perform a more robust derivation that gives us the interval:

$$EEP_P \in \{\max[0, P_A - C_A - PV(X) - PV(D^e) + S_0 - \lambda_A], P_A - C_A - PV(X) + S_0 + \lambda_A\}$$

We examine the early exercise premium (EEP) of American SPY puts by bounding the premium between theoretical lower and upper limits derived from put–call parity with adjustments for dividends, transaction costs, and bid–ask spreads. Because the premium cannot be observed directly, it is treated as a latent variable within these bounds. Rather than relying solely on the midpoint of the interval, we apply a Monte Carlo simulation that repeatedly samples latent realizations of ln(EEP) between the bounds. To reduce simulation variance, we implement antithetic variates by pairing each uniform draw with its complement. This procedure produces thousands of feasible paths for the unobserved premium.

For each simulated series, we estimate an ARMA(3,0,3) model with robust standard errors, regressing ln(EEP) on time to maturity (T) and the implied volatility of SPY.

The results are stable across 1,000 imputations. The coefficient on time to maturity is strongly positive (average $\approx$ 10.65, 95% Monte Carlo interval: 10.22–11.00), indicating that the value of early exercise rises as maturity increases, consistent with theoretical predictions. The coefficient on implied volatility is also positive (average $\approx$ 0.295, 95% interval: 0.264–0.331), confirming that greater market-expected volatility increases the option holder's incentive to exercise early. These findings hold under repeated simulation, demonstrating that the key relationships are not artifacts of midpoint approximation but instead reflect persistent pricing behavior in SPY options.

This specification, therefore, captures the main theoretical drivers of the early exercise premium under realistic market conditions, while accounting for both unobservable variables and empirical frictions.

3 Empirical Analysis and Results

3.1 Construction of the Dependent Variable

3.1.1 Data Collection for Estimating EEP_P

We cannot directly observe the early exercise premium of an American put option (EEP$_P$) since an exact calculation requires the prices of American and European options, and exchanges primarily list American-style options. However, leveraging established pricing relationships, we can arrive at an estimate of EEP$_P$ as a function of collectible data, that is, American put and call prices, spot market prices, bid-ask spreads, the riskless rate, strike prices, ex-dividend dates, dividend yield, and option maturity.

To calculate EEP_P, we collected data from the SPDR S&P 500 ETF (SPY), an American put and call option with SPDR as the underlying (both have 560 strikes and expire on 06/20/2025), and the United States Generic Government 3-Month Treasury Bill yield through the Bloomberg Terminal.

The summary of the raw Bloomberg Terminal data is as follows:

Variables Used For Calculating EEP$_P$						
Variable	Obs	Mean	Std. Dev.	Min	Max	Notation
Date	250	23669.77	105.6869	23489	23853	$\cdots$
PX_BID_P_SPY	250	23.44168	12.74852	6.07	65.56	$P_{A,bid}$
PX_ASK_P_SPY	250	23.9718	13.33779	6.11	68.53	$P_{A,ask}$
PX_MID_P_SPY	250	23.70856	13.03962	6.09	67.05	$P_A(S_0', T, X)$
PX_BID_C_SPY	250	40.93056	15.45545	5.15	68.5	$C_{A,bid}$
PX_ASK_C_SPY	250	41.83	16.02096	5.23	71.15	$C_{A,ask}$
PX_MID_C_SPY	250	41.38476	15.73257	5.19	69.83	$C_A(S_0', T, X)$
PX_BID_S_SPY	250	564.3715	29.34952	496.75	612.9	$S_{0,bid}$
PX_ASK_S_SPY	250	564.3902	29.34138	497	612.91	$S_{0,ask}$
PX_MID_S_SPY	250	564.3796	29.34961	496.76	612.905	S_0
Dvd_Yld_S_SPY	250	1.256887	0.0623294	1.1487	1.4258	$\cdots$
MID_USGG3M	250	4.76442	0.45466	4.2425	5.4093	$\cdots$

There are a few other data points that are important for estimating EEP_P.

The five ex-dividend dates for SPY over the lifespan of the options:

- 6/21/24, 9/20/24, 12/20/24, 3/21/25, 6/20/25

The options expiration date:

- 6/20/25

And finally, the options strike:

- X = \$560

3.1.2 Calculation of Present Value of Expected Dividends and Strike

The present value of expected dividends and strike must be used to estimate EEP_P. However, like EEP_P, these present values cannot be observed directly. Still, we can get a close approximation.

The 3 month annualized t-bill yield rate is a consistent short-term, historically risk-less, interest rate. So, the annualized rate can be used as a proxy for the continuous risk-free rate.

$$r_e \approx \ln\left(1 + \frac{\text{PX_MID_USGG3M}}{100}\right)$$

Calculating the annualized time until expiration (T) and time until each ex-dividend date (t_j) is also straightforward.

$$T = \frac{\text{Days Until Expiration}}{365}, t_j = \frac{\text{Days Until Ex-Dividend Date}}{365}$$

Given r_e and T, we can now discount the strike price using continuous compounding:

$$\text{Present Value of Strike} = X \cdot \exp(-r_e T)$$

SPY has consistent **quarterly** ex-dividends dates without drastic fluctuations. The present value of dividends depends on the expected amount, not the actual amount. However, we can make a close approximation of the changes in SPY dividend expectations using the daily annualized dividend yield.

$$D_j^e = \frac{\text{Dvd_Yld_S_SPY} \cdot S_0}{4 \cdot 100}$$

Now we have all of the data to calculate the present value of expected dividends $= \Sigma_{j=1}^{N} D_j^e \cdot \exp(-r_e t_j)$.

3.1.3 Estimating Theoretical EEP$_P$

To estimate EEP$_P$, we must manipulate the European Put-Call-Parity. The European Put-Call-Parity of a dividend-paying stock, where $S_0' = S_0 - \Sigma_{j=1}^{N} D_j^e \cdot \exp(-r_e t_j)$, can be written as followed:

$$P_E(S_0', T, X) + S_0' = C_E(S_0', T, X) + X \cdot \exp(-r_e T)$$

By rearranging the Put-Call-Parity, we can isolate the price of a European put:

$$P_E(S_0', T, X) + S_0 - \Sigma_{j=1}^{N} D_j^e \cdot \exp(-r_e t_j) = C_E(S_0', T, X) + X \cdot \exp(-r_e T)$$
$$P_E(S_0', T, X) + S_0 = C_E(S_0', T, X) + X \cdot \exp(-r_e T) + \Sigma_{j=1}^{N} D_j^e \cdot \exp(-r_e t_j)$$

$$\boldsymbol{P_E(S_0', T, X) = C_E(S_0', T, X) + X \cdot \exp(-r_e T) + \Sigma_{j=1}^{N} D_j^e \cdot \exp(-r_e t_j) - S_0} \quad (1)$$

Thus, we can calculate the European put price given the European call price, the present value of the strike and expected dividends, and today's spot price of the underlying.

For clarity, let us simplify the notation,

$$\begin{aligned} P_E &\equiv P_E(S_0', T, X) \\ C_E &\equiv C_E(S_0', T, X) \\ PV(X) &\equiv X \cdot \exp(-r_e T) \end{aligned} \qquad \begin{aligned} P_A &\equiv P_A(S_0', T, X) \\ C_A &\equiv C_A(S_0', T, X) \\ PV(D^e) &\equiv \Sigma_{j=1}^{N} D_j^e \cdot \exp(-r_e t_j) \end{aligned}$$

Because not many European options are publicly traded, we must adjust the equation in terms of American options. The holder of an American option, instead of having to wait until expiration, can exercise the option at any time before the expiration date. An American option is a European option with more flexibility. Therefore, given the same spot price, maturity date, and strike price, an American option must be priced at least as high as a European option:

$$\begin{aligned} P_A &\geq P_E \\ \therefore P_A &= P_E + EEP_P \\ EEP_P &= P_A - P_E \end{aligned} \qquad \begin{aligned} C_A &\geq C_E \\ \therefore C_A &= C_E + EEP_C \\ C_E &= C_A - EEP_C \end{aligned}$$

Given that $T > 0$, the profit of selling the call option must be greater than that obtained by exercising —absent dividend payments. The only reason an owner would wish to exercise a call would be before an ex-dividend date to receive that and the future dividends that they otherwise could not get. Thus, the maximum benefit (and hence the maximum premium paid) of exercising a call before maturity is the underlying's present value of future dividends.

$$\begin{aligned} \text{So,} \quad EEP_C &\in [0, PV(D^e)] \\ \therefore C_E &\in [C_A - PV(D^e), C_A] \end{aligned}$$

Returning to the Put-Call-Parity for European options:

$$\begin{aligned} P_E + S_0 &= C_E + PV(X) + PV(D^e) \\ P_E &= C_E + PV(X) + PV(D^e) - S_0 \\ EEP_P &= P_A - P_E \end{aligned}$$

Substituting for P_E from equation (1):

$$\begin{aligned} EEP_P &= P_A - [C_E + PV(X) + PV(D^e) - S_0] \\ &= P_A - C_E - PV(X) - PV(D^e) + S_0 \end{aligned}$$

After also substituting for C_E we get a measurable range for an American put's theoretical early exercise premium.

$$\boldsymbol{EEP_P} \in \{\max[0, \boldsymbol{P_A - C_A - PV(X) - PV(D^e) + S_0}], \boldsymbol{P_A - C_A - PV(X) + S_0}\} \quad (2)$$

We have data for every variable in Equation (2). However, Equation (2) is incomplete because we have ignored transaction costs.

3.1.4 Estimating Practical EEP$_P$

The put call parity relies on arbitrage and the efficient market assumption. The efficient market assumption is fairly accurate, as any apparent arbitrage can be immediately exploited through algorithmic trading. However, to take advantage of an arbitrage opportunity for European put, the profit must exceed the transaction cost. Equation (2) ignores those transaction costs.

The largest transaction cost is the put-call spread. When purchasing a put or call, market participants can only purchase above the theoretical fair value —the bid price. Likewise, when selling, participants can only sell below the theoretical fair value —the ask price. Thus, we can refine Equation (2) by accounting for every spread's transaction cost,

$$\lambda_E = \frac{1}{2}[(P_{E,ask} - P_{E,bid}) + (C_{E,ask} - C_{E,bid}) + (S_{0,ask} - S_{0,bid})]$$

By subtracting the total transaction costs from $\min[EEP_p]$ and adding the costs to the maximum $\max[EEP_p]$, we can account for the practical limitations of the put-call parity. However, again, we only have access to American bid-ask spreads. So, we must use

$$\lambda_A = \frac{1}{2}[(P_{A,ask} - P_{A,bid}) + (C_{A,ask} - C_{A,bid}) + (S_{0,ask} - S_{0,bid})]$$

The bid-ask spreads for the American options are slightly higher than those of the European options. As such, we are conservative in our estimation when we add American transaction costs to the range. We are slightly widening the range more than necessary. Including transaction costs in the equation, we get our final equation.

$$\boldsymbol{EEP_P} \in \{\max[0, \boldsymbol{P_A - C_A - PV(X) - PV(D^e) + S_0 - \lambda_A}], \boldsymbol{P_A - C_A - PV(X) + S_0 + \lambda_A}\} \quad (3)$$

3.1.5 Automation of Bound Calculations

To ensure consistency and scalability, we automated the calculation of EEP_{min} and EEP_{max} directly from their mathematical definitions. Instead of computing bounds manually, our pipeline parses option chain data (in CSV format) and applies the no-arbitrage relationships programmatically to generate valid intervals for each observation.

This automation provides several advantages:

- Eliminates human error in repetitive calculations.

- Ensures that every data point is checked against the same mathematical logic.
- Enables fast processing of large datasets (25M+ synthetic option pricing rows in stress tests).

This procedure not only streamlined our analysis but also made the framework reusable for future studies of option pricing dynamics.

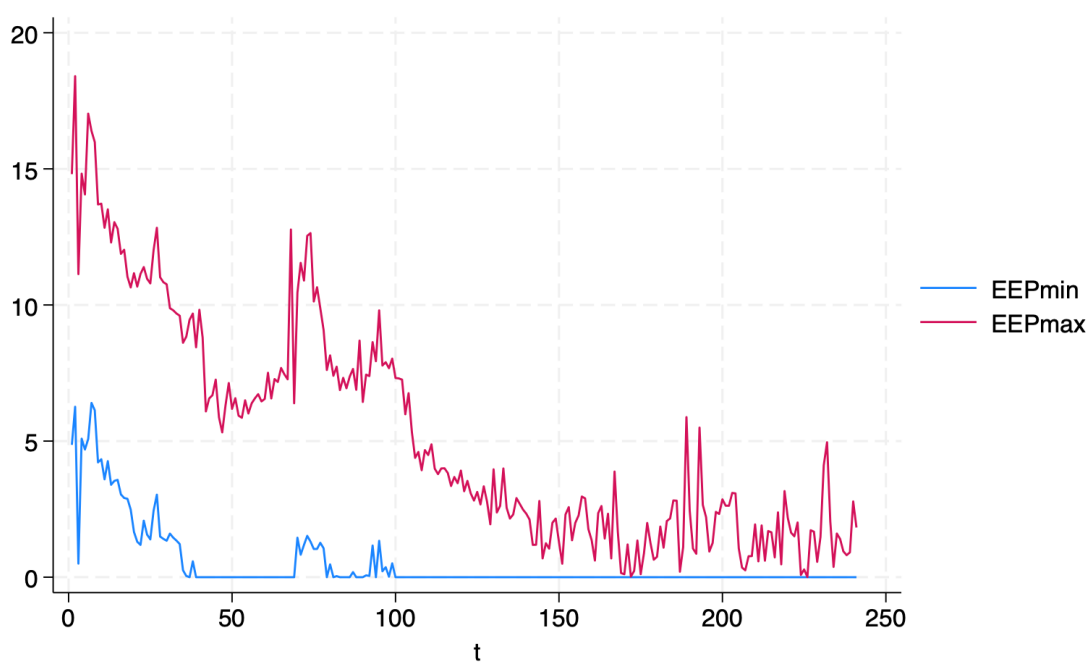
3.1.6 EEP$_P$ Data Limitations

Although the range for EEP_P is conservative, the collected data have one noticeable flaw. Bloomberg Terminal offers closing options and spot prices; however, options on SPY trade until 4:15 p.m. ET while the SPY ETF closes at 4:00 p.m. ET. The time difference can create price misalignment, resulting in "false flags" where it appears that arbitrage is possible in the data.

Although arbitrage opportunities may present themselves (where EEPmax < 0), arbitrage is not actionable and only represents strong market changes in the last 15 minutes before options close. These "false flags" particularly appear in the data after April 2nd, when Trump's Liberation Day announcement caused fast changes in the spot and options market.

So, whenever EEPmax < 0, we will drop the data point. In the 250 observations, this happens seven times.

After dropping those variables, we get a EEP$_P$ interval that looks like this:



3.2 Monte Carlo Approach

We now have the theoretical bounds for the premium:

$$EEP_P \in \left[EEP_{\min},\, EEP_{\max}\right],$$

The true value of EEP_P lies somewhere between these bounds, and is not directly observable for regression. To address this, we employ a Monte Carlo simulation framework that repeatedly samples latent realizations of the unobserved premium within its feasible interval. Specifically, we work in log-space to stabilize variance and ensure positivity. For each draw m, we generate:

$$\ell EEP^{(m)} = \ell Lb + u \cdot (\ell Ub - \ell Lb), \quad u \sim U(0,1),$$

where $\ell Lb = \ln(EEP_{\min})$ is the log of the lower bound, $\ell Ub = \ln(EEP_{\max})$ is the log of the upper bound, and u is a random variable drawn from the uniform distribution $U(0,1)$. This produces a log-uniform distribution of simulated values consistent with the theoretical bounds.

To improve efficiency and reduce variance, we also implement *antithetic variates*, pairing each draw u with its complement $u' = 1 - u$:

$$\ell EEP^{(m,1)} = \ell Lb + u \cdot (\ell Ub - \ell Lb), \quad \ell EEP^{(m,2)} = \ell Lb + (1-u) \cdot (\ell Ub - \ell Lb).$$

This ensures symmetry in the sampling process and stabilizes the distribution of simulated coefficients.

Regression Specification

For each imputed series of $\ln(EEP_P)$, we estimate the following model:

$$\ln(EEP_P) = \beta_0 + \beta_T T + \beta_{ivM}\, ivM + \varepsilon_t, \quad \varepsilon_t \sim ARMA(3,0,3),$$

where T is the time-to-maturity in years until expiration, ivM is the implied volatility of SPY at-the-money options (sourced from Bloomberg's 3-month implied volatility series), and ε_t represents the residuals modeled with an ARMA(3,0,3) process to capture autocorrelation.

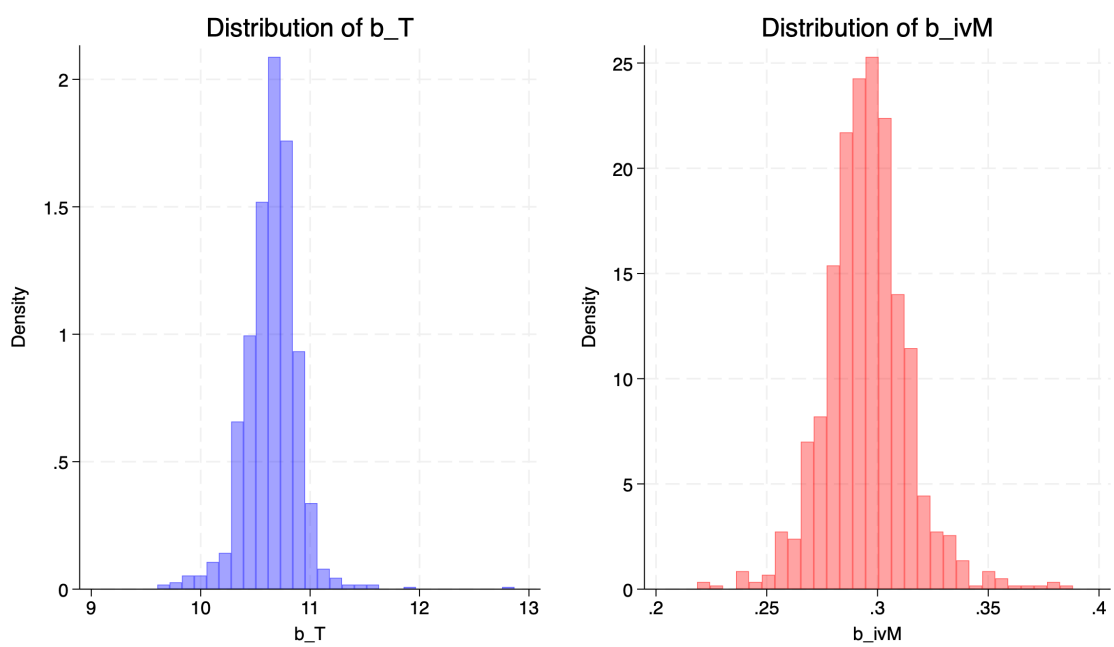
Why ARMA? Preliminary residual diagnostics indicated significant autocorrelation, which if left uncorrected, would bias inference. After testing several specifications, the ARMA(3,0,3) model provided the best balance between fit and parsimony, effectively removing serial dependence and producing white-noise residuals. Including autoregressive (AR) and moving-average (MA) terms allows us to isolate the true effects of T and ivM rather than absorbing them into spurious temporal patterns.

Why not difference the series? Unit root tests (e.g., ADF) suggested nonstationarity in the raw $\ln(EEP_P)$. However, differencing would strip the data of economically meaningful long-run variation and complicate interpretation of coefficients. Instead, we treat the near-unit-root behavior with ARMA error terms, which control for persistence without discarding the level relationship between EEP_P, time-to-maturity, and implied volatility. Importantly, the estimated AR and MA lag coefficients always sum to less than one, confirming that the ARMA specification is stable and that the process reverts toward equilibrium. This choice preserves the interpretation that a negative coefficient on T reflects the theoretical decline in the premium as expiration approaches.

3.2.1 Results

We run $M = 1000$ Monte Carlo replications, yielding 1000 coefficient estimates for both β_T and β_{ivM}. The resulting distributions are approximately normal and tightly centered:

- For **time-to-maturity** (T), the coefficients are strongly negative across simulations, confirming that EEP_P decreases as expiration approaches.
- For **implied volatility** (ivM), the coefficients are consistently positive, reinforcing that greater expected volatility increases the option holder's incentive to exercise early.



Monte Carlo Regression Results (1000 Replications)					
Variable	Obs	Mean	Std. Dev.	Min	Max
b_T	1000	10.6513	0.2417	9.6037	12.8673
b_{ivM}	1000	0.2952	0.0188	0.2186	0.3882

The Monte Carlo simulations yield stable and statistically significant coefficients. For the time-to-maturity coefficient β_T, the mean estimate is 10.65 with a standard deviation of 0.24. This produces an average t-statistic of roughly 44.1, implying $p < 0.001$.

For the implied volatility coefficient β_{ivM}, the mean estimate is 0.295 with a standard deviation of 0.019. This corresponds to an average t-statistic of about 15.7, also with $p < 0.001$.

Both coefficients remain highly significant across the 1,000 simulated replications, confirming that the observed relationships between EEP_P, time-to-maturity, and implied volatility are robust and not artifacts of a single imputation.

3.2.2 Interpretation of Coefficients

The Monte Carlo regression results allow us to translate the estimated coefficients of time-to-maturity and implied volatility into their economic meaning for the log of the early exercise premium, $\ln(EEP_P)$:

- **Time-to-maturity** (T): The mean coefficient on T is 10.65. Interpreted literally, this means that a one-year increase in time-to-maturity raises the early exercise premium by approximately
$$(e^{10.65} - 1) \times 100 \approx 4.23 \times 10^6\%,$$
an extremely large effect. This scale issue arises because T is measured in years; thus, a full one-year increase is rarely observed in the data. Using a finer unit, such as months, gives more practical results. For example, dividing the coefficient by 12, we obtain a monthly effect of $\beta_T \approx 0.888$, which translates to about $(e^{0.888} - 1) \times 100 \approx 142.9\%$ per month. This confirms the theoretical prediction: as time-to-maturity increases, the premium rises sharply, while as expiration approaches the opportunity to benefit from volatility disappears and the premium declines.

- **Implied volatility** (ivM): The mean coefficient on implied volatility is 0.295. This implies that a one-percent increase in implied volatility (measured at the 100% moneyness level for SPY) increases the early exercise premium by approximately
$$(e^{0.295} - 1) \times 100 \approx 34.3\%.$$
Thus, higher expected market volatility increases the likelihood that the put finishes deeply in the money before expiration, making early exercise more attractive and raising the premium.

- **Constant term**: The constant reflects the baseline of $\ln(EEP_P)$ when both T and ivM are set to zero. While it has no stand-alone economic interpretation, it anchors the regression and ensures that the fitted values are properly scaled relative to the observed data.

Overall, these interpretations show that both time-to-maturity and implied volatility play economically meaningful roles in determining the size of the early exercise premium, with time effects particularly strong once scaled to realistic horizons.

4 Conclusion

This paper estimated the early exercise premium (EEP) of an American put option on the SPDR S&P 500 ETF by combining theoretical bounds with simulation-based methods. Using bid–ask quotes, dividend yields, risk-free discounting, and spot prices, we constructed bounds for the premium based on put–call parity with frictions. Because the premium is not directly observable, we employed a Monte Carlo framework to draw latent realizations of $\ln(EEP_P)$ between its theoretical lower and upper bounds. This approach allowed us to quantify the premium's sensitivity to time-to-maturity and implied volatility while accounting for uncertainty in the precise value of the premium.

The results are consistent with option pricing theory. Time-to-maturity has a strongly positive effect on the early exercise premium: longer horizons significantly increase the value of waiting to exercise. Implied volatility also raises the premium, reflecting that greater market uncertainty enhances the likelihood of early exercise becoming optimal. Together, these effects confirm that the premium is driven by both the length of time available for favorable price movements and the market's expectations of future volatility.

We also incorporated ARMA error corrections to address autocorrelation in the time-series of simulated premiums. Although unit root tests suggested nonstationarity, differencing would have discarded long-run variation. By instead modeling persistence through ARMA(3,0,3), we preserved the level relationship between the premium, time, and volatility while ensuring stability (lag sums consistently below unity). This produced well-behaved residuals and statistically robust estimates across 1,000 replications.

Overall, this study shows that the early exercise premium is economically meaningful and empirically robust, even when only bounds are observable. By integrating bounds construction with Monte Carlo methods, we demonstrate how statistical simulation can recover latent option values under realistic market frictions, providing a bridge between theoretical pricing conditions and practical market data.